

ADR: GEO Database Migration Target and SQL Platform Direction

Status: PROPOSED

Task URL: [RMASUP-2134](#)

Date: 2026-05-22

1. Current Context

Area	Details
Task goal	Give the client setup guidance and best practices for the new modernized SQL server.
Change trigger	The current GEO database is running on SQL Server 2017 on soon-to-be unsupported operating systems, and the current GEO server has recurring performance issues.
Relevant constraints	• GEO moves first. • RMA may move later to the same target. • SQL Server feature compatibility is important. • Some procedures use local Windows disk. • SSRS is part of the current setup. • Final cost still needs validation. • Final migration scope still depends on the assessment report.
Existing pattern or baseline	The current solution uses server-based SQL Server infrastructure, and SQL Server plus reporting are tied to Windows Server management overhead.

2. Chosen Direction

- Recommend Azure SQL Managed Instance as the target database platform.
- Treat GEO migration as phase 1 and keep the path open for a later RMA migration into the same environment.
- Keep SSRS engine hosting as a separate workstream, and do not treat this ADR as final cost approval, final sizing, or full migration approval.

3. Decision Scope

Scope Type	Details	Client-friendly explanation
In scope	Select the preferred platform direction for the new SQL environment.	This ADR decides the recommended target direction, not every technical detail of the final migration.
Out of scope	• Final cost approval. • Exact SKU sizing. • Exact migration runbook. • Full component inventory. • Final SSRS engine hosting design.	These items still depend on the migration assessment and later planning work.
Assumptions	• The client needs a recommendation now. • Detailed validation will come after access and assessment. • GEO is the first migration target.	The team needs a clear recommendation first, while keeping room for validation before final execution.

4. Decision Reasons

Main decision

- Azure SQL Managed Instance was chosen because it best matches the current SQL Server feature set, reduces rewrite risk, and provides a better long-term operating model than the likely alternatives.
- It is also the most practical balance between compatibility, modernization, and future GEO plus RMA consolidation.

Why it fits the current architecture or team direction?

Category	Reason	Client-friendly explanation
SQL compatibility	It supports SQL Agent, linked servers, cross-database queries, and other SQL Server features used in the current environment.	This lowers migration risk and reduces the amount of application or database rewrite work.
Reporting support	It can host SSRS catalog databases, which fits the current reporting setup better than	The reporting data layer can move with the rest of the SQL workload, even if the

	Azure SQL Database.	SSRS engine remains on a VM.
Operating model	It removes the Windows Server layer and reduces overhead around patching, backups, and built-in high availability.	This simplifies long-term maintenance and reduces the amount of infrastructure the team needs to manage directly.

Why it is better than the likely alternatives for this case?

Summary	Reason	Client-friendly explanation
Better than Azure SQL DB	Azure SQL Database would likely require more rewrite work and has a known SSRS catalog limitation.	It is less suitable for the current workload because it creates more migration risk and feature gaps.
Better than staying on VM	Managed Instance modernizes the main database workload while avoiding the larger Windows and SQL Server VM management burden.	The heavier SQL workload moves to a managed platform, while only a smaller and simpler reporting VM remains.
Better for long-term path	It supports GEO migration now while leaving room for a later RMA move into the same managed environment.	This creates a cleaner long-term consolidation path without forcing all migration work into one phase.

5. Options Considered

Option	Benefits	Tradeoffs	Client-friendly explanation
Option 1 — Azure SQL Managed Instance	• High SQL Server compatibility. • Supports SQL Agent, linked servers, and cross-database queries. • Supports SSRS catalog databases. • Reduces Windows Server overhead. • Includes high availability and backups. • Supports GEO-first migration and possible later RMA migration.	• Still needs assessment before cost and sizing are trusted. • Some file-based procedures must be refactored. • SSRS engine still needs separate hosting, typically on a Windows VM.	This is the best fit for the current system because it keeps strong SQL Server compatibility while improving the long-term operating model.
Option 2 — Azure SQL	• Managed PaaS	• Lower SQL Server	This is simpler as a pure

Database	database service. • Less infrastructure to manage than a VM.	compatibility. • Cannot host SSRS catalog databases. • Likely needs more rewrite work.	platform service, but it creates more migration risk for the current system.
Option 3 — New SQL Server on VM	• High short-term compatibility. • Easier lift-and-shift path.	• Keeps Windows Server overhead. • Keeps more maintenance burden. • Gives less modernization benefit.	This is safer short term, but weaker as a long-term platform choice.

6. Change Impact

Area	Impact
Code impact	• Procedures that read or write files on local Windows disk must be refactored. • The cleanest approach is to move file handling into the app layer.
Data / schema impact	• GEO database moves first. • SSRS catalog databases can move to Azure SQL Managed Instance. • Full compatibility still needs assessment. • The full final list of databases and services still depends on the assessment report.
Security impact	Azure access and permissions must be coordinated to run the assessment and manage the target environment.
Performance impact	The target should improve operations and reliability, but final sizing depends on workload data and assessment output.
Operational impact	• The team needs Azure access or help from the client team to run the assessment. • The SSRS engine remains on a smaller VM for now. • The assessment may need about one week of performance data to support better SKU recommendations.
Testing impact	• Migration needs compatibility checks. • Migration needs workload checks. • Migration needs reporting checks.

- Migration needs post-migration verification.

7. Expected Outcomes

Area	Details	Client-friendly explanation
Benefits	• Clear platform direction now. • Lower long-term infrastructure overhead. • Better SQL Server compatibility than Azure SQL Database. • Room for future GEO and RMA consolidation.	The team can move forward with planning now while still keeping a practical long-term direction.
Tradeoffs	• Some decisions remain open. • Azure access is still a blocker. • Some refactoring may be needed. • The reporting engine remains on a VM in the short term.	This is a strong direction, but it still depends on assessment results and a few follow-up decisions.
Follow-up work	• Get the first assessment report. • Review cost estimates. • Review SKU recommendation. • Collect about one week of performance data if possible. • Align the plan with Dmitry before wider sharing. • Confirm the SSRS plan. • Prepare the migration runbook.	The next step is validation and planning, not immediate full execution.
Risks	• Hidden compatibility issues may appear in the assessment. • Access delays may slow planning. • Actual cost may change after real sizing data is available.	The recommendation is strong, but it still needs assessment data to reduce surprise risks.

8. Pending Items

Type	Details
Pending validations	• Migration assessment report. • Compatibility blockers. • Affected objects. • Full component inventory. • SKU recommendation. • Workload-based sizing. • Cost estimates.
Pending dependencies	• Azure subscription or VM-level access to register the SQL IaaS Agent extension and run the assessment. • Or the client team running it and sharing the report. • Alignment with Dmitry before wider team sharing.
Deferred decisions	• Final SSRS engine hosting approach. • Final migration runbook details. • Final decision on GEO and RMA sharing one managed instance. • Possible future move from SSRS to another reporting approach if needed.

9. Supporting Evidence

- **Task evidence:** The Jira task asks for setup guidelines and best practices for the new SQL server, with GEO moving first and RMA likely moving later to reduce the cost and effort of managing two servers at the same time.
- **Plan evidence:** Comments recommend Azure SQL Managed Instance over Azure SQL Database or staying on a VM because of better SQL Server compatibility, SSRS catalog support, removal of Windows Server overhead, and built-in high availability and backups.
- **Review evidence:** Comments state that a migration assessment should reveal compatibility blockers, affected objects, SKU recommendation, and cost estimates. They also note that Azure access is needed to run the assessment, and that some procedures depend on local Windows disk.

External references

- [SQL Server end of support options](#) — confirms SQL 2017 extended support ends 2027; recommends MI as first migration option.
- [What is Azure SQL Managed Instance?](#) — confirms near-100% SQL Server compatibility, SQL Agent, linked servers, cross-database queries, automated patching/backups/HA, SSRS catalog hosting.
- [Features comparison: SQL DB vs SQL MI](#) — confirms SQL DB lacks SQL Agent, linked servers, cross-database queries, and SSRS support that the current GEO system depends on.
- [SQL MI — Business Intelligence section](#) — confirms MI can host SSRS catalog databases while SSRS engine runs on a VM.
- [End of support — Azure VM vs MI comparison](#) — confirms VM option still requires managing both SQL Server and OS; MI option is fully managed with evergreen features.